

MARC SANCHEZ RIVERA

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PROFESSIONAL SUMMARY

I'm a Research Engineer with 3 years of hands-on experience in machine learning systems and data infrastructure. I've worked on projects ranging from NLP model optimization to distributed computing pipelines, mostly in startups where wearing multiple hats is the norm. I'm particularly interested in making complex systems more efficient and spending a lot of my time understanding where bottlenecks actually are rather than assuming. Based in Barcelona, I'm looking for roles where I can deepen my research work while contributing to real product impact.

WORK EXPERIENCE

Research Engineer

DeepMind Barcelona Labs

Barcelona, Spain

deepmind.com

January 2023 - Present

Developed and optimized transformer-based models for sequence-to-sequence tasks, reducing inference latency by 35 percent through quantization and pruning techniques. Implemented distributed training pipeline using PyTorch Lightning across 8 GPU nodes, cutting model training time from 72 hours to 18 hours. Conducted ablation studies on attention mechanisms and contributed findings to two internal research papers currently under peer review. Mentored two junior engineers on best practices for experiment tracking and reproducible machine learning workflows.

Machine Learning Engineer

NexusAI Solutions

nexusai.io

Barcelona, Spain

June 2021 - December 2022

Built data preprocessing and feature engineering pipeline handling 500 million daily events using Apache Spark and Python, improving data quality metrics by 28 percent. Collaborated with product team to A/B test three different recommendation algorithms, with the winning variant increasing user engagement by 12 percent. Designed monitoring dashboards in Grafana and Prometheus to track model performance drift in production environments. Established reproducibility standards for the team including version control practices for datasets and model checkpoints.

Junior Research Engineer

TechVision Research Group

techvision.net

Barcelona, Spain

September 2020 - May 2021

Implemented computer vision algorithms for document classification achieving 94 percent accuracy on internal benchmark dataset. Conducted literature reviews on recent advances in CNNs and documented findings in technical reports shared across the research team. Contributed to documentation and refactoring of legacy codebase, reducing technical debt and improving code maintainability. Participated in weekly research seminars

presenting papers and discussing emerging trends in deep learning.

EDUCATION

Master of Science in Computer Science with specialization in Artificial Intelligence

Universitat Politècnica de Catalunya (UPC)

Barcelona, Spain

2019 - 2021

Bachelor of Science in Computer Science

Universitat Autònoma de Barcelona (UAB)

Barcelona, Spain

2015 - 2019

SKILLS

Machine Learning and AI: PyTorch, TensorFlow, Scikit-learn, XGBoost, transformer models, deep learning, NLP, computer vision, model optimization

Data Engineering: Apache Spark, SQL, data pipelines, ETL, feature engineering, data quality assurance

Tools and Platforms: Jupyter Notebook, Git, Docker, Kubernetes basics, Weights and Biases, Hugging Face, MLflow

Programming Languages: Python, Java, SQL, Bash

Infrastructure and Monitoring: Grafana, Prometheus, distributed training, GPU optimization

Research: Experiment design, ablation studies, literature review, scientific writing